

Healthcare Operations & Revenue Cycle Analytics

1. Project Overview and Objective

This project involves cleaning, transforming, and analysing raw healthcare data using **Excel** and creating an interactive **Power BI dashboard** to derive meaningful business and clinical insights.

The main objective is to demonstrate data pre-processing techniques using Excel and an interactive Power BI dashboard visualization to evaluate patient demographics, encounter volumes, revenue realization across medical specialties, and insurance payment realization rates to support informed healthcare management decisions

2. Data Sources

- Source Description and Timeline: Healthcare Enterprise Administrative Records (Patients Master, Encounters Log, Vitals, Diagnoses, Procedures, Charges, and Claims) spanning 2022 to 2026.
- Domain: Healthcare Analytics / Clinical Operations & Revenue Cycle Management

3. Problem Statement

- To analyse financial realization performance across provider specialties to identify top revenue-generating medical departments and billing gaps.
- To evaluate clinical encounter patterns across patient demographics (paediatric vs. young adult) and encounter types (Telehealth, Office Visit, Emergency, Hospital Admission).
- To monitor monthly revenue trends and patient visit volumes to optimize clinical scheduling and resource allocation.
- To analyse cost contribution categories (High Cost vs. Standard) and insurance claim paid ratios to optimize revenue cycle efficiency.

4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Encounter ID	String / Text	Unique identifier for each patient visit/encounter
Patient ID	String / Text	Unique identifier for individual patients
Age Category	Categorical	Demographic age classification (Paediatric, Young Adult)
Provider Speciality	Categorical	Medical department (Cardiology, Gastroenterology, Oncology, etc.)
Charge Amount	Numeric (Decimal)	Total billed amount for the encounter (\$)

5. Tools & Technologies:

- **Excel:** Data cleaning, transformation, relational lookup modelling, calculated fields, and Pivot Tables.
- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

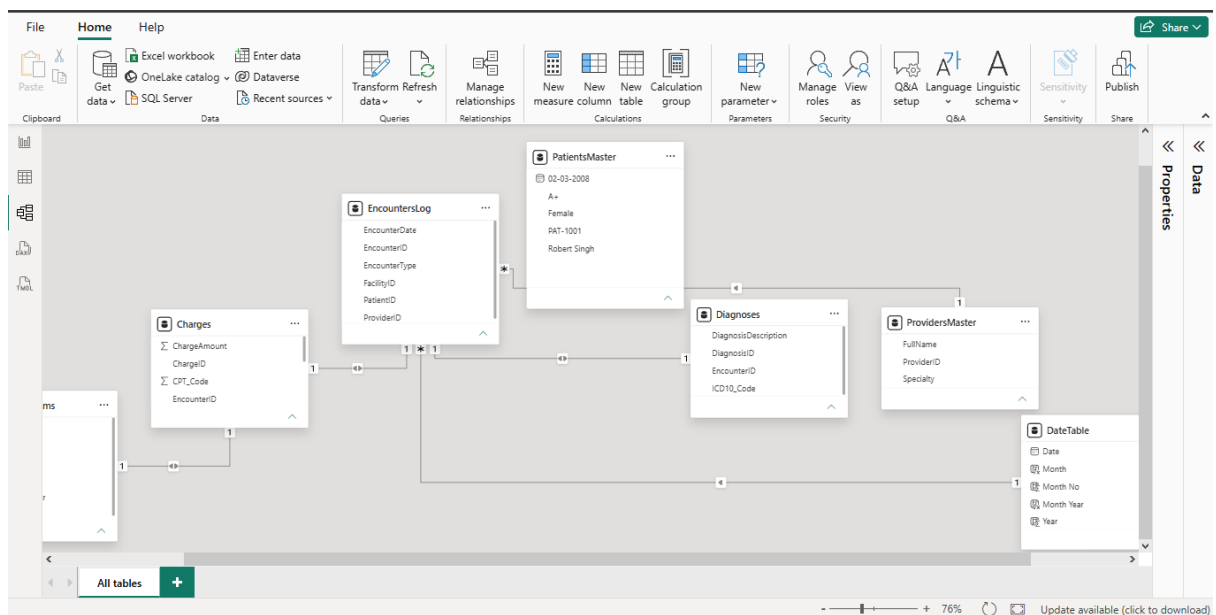
6. Data Pre-Processing (Excel / Power Query):

- **Data Cleaning & Transformation:** Cleaned 2,766 encounter records, removed duplicate entries, formatted currency fields, and unified ICD-10 diagnosis descriptions.
- **Filtering & Sorting:** Filtered out unbilled charges and standardized date attributes into Year Month formats.
- **Pivot Tables:** Generated Pivot Tables for summary analysis across Revenue Realization, Clinical Demographics, Monthly Patient Volume, and Revenue Contribution.
- **Convert the data into Fact and Dimension Table:** Modelled relational lookup architecture connecting dimension tables (Patients Master, Providers Master, Patient Contacts) with transaction logs (Encounters Log, Charges, Insurance Claims) into a unified Master Data fact sheet.

7. Data Modelling and DAX (Power BI):

- **Data Model:** Established 1-to-Many relationships between Dimension Master tables (Patient ID, Provider ID) and Fact tables (Encounter ID, Charge ID, Claim ID).

DATA MODELLING:



- **Calculated Columns & DAX Measures:**
 - **Total Billed Amount:** SUM ('Master Data'[Charge Amount])
 - **Total Paid Revenue:** SUM ('Master Data'[Claim Paid Amount])
 - **Payment Realization Rate:** DIVIDE ([Total Paid Revenue], [Total Billed Amount], 0)

8. Analysis and Visualizations (Power BI):

Dashboard Features:

- **Multiple Visualizations based on problem statement:**
 - **KPI Cards:** Total Billed (\$775.56K), Total Collected (\$558.00K), Total Visits (2,766), Unique Patients (500).
 - **Donut Chart / Tree map:** Revenue Contribution split by Cost Bracket (High Cost: \$192.71K vs. Standard: \$365.29K).
 - **Line Chart:** Monthly Revenue Trends & Patient Visit Volume across time.
- **Interactive Layout:** Equipped with slicers for **Encounter Type**, **Provider Specialty**, and **Date Range**.

- **Bookmarks & Navigation:** Custom reset buttons and clear titles on all visuals.

CARD VISUAL:

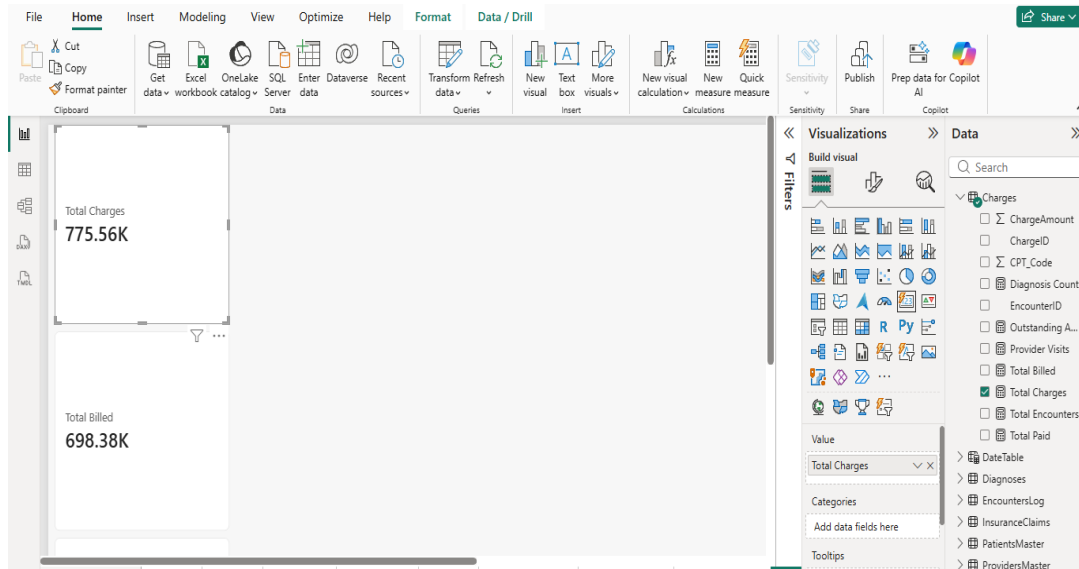
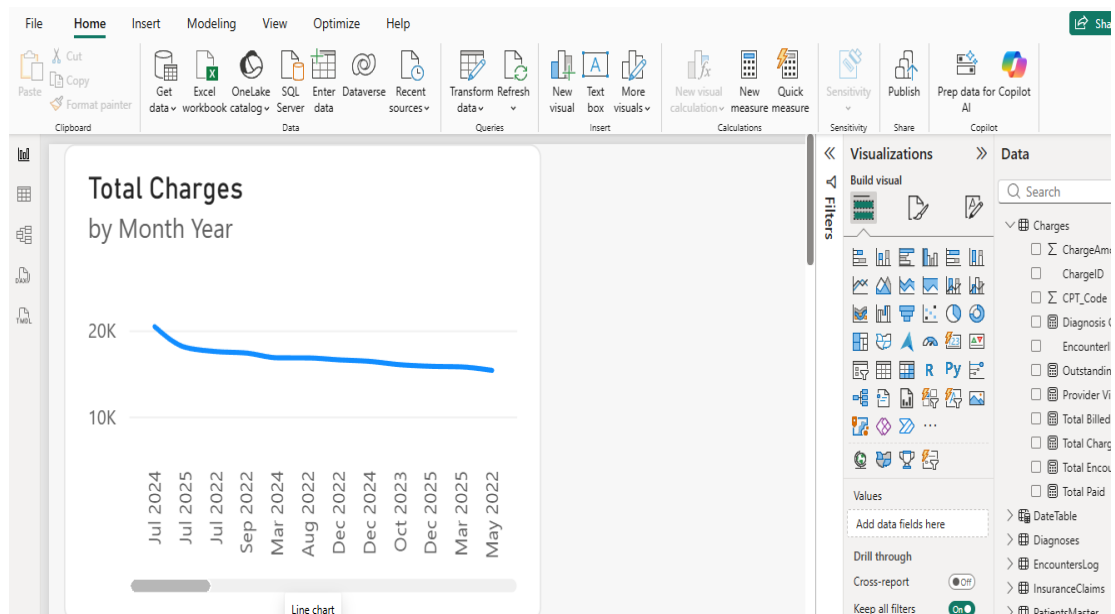


CHART VISUAL:



9. Insights & Conclusions :

- Key Findings:
 - **Volume Distribution:** 2,766 total encounters spread across Telehealth (736), Office Visits (684), Emergency (675), and Hospital Admissions (671).
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- **Provide the analysis insights:**
 - **Descriptive:** Gastroenterology (\$171.18K) and Cardiology (\$142.95K) account for over 56% of total revenue collected.
 - **Diagnostic:** Revenue realization gaps are primarily driven by insurance claims adjustment margins in high-cost procedure categories.
 - **Predictive:** Emergency and Hospital Admissions correlate directly with High-Cost bracket claims (\$192.71K total).
 - **Prescriptive:** Focus clinical resource scheduling toward Gastroenterology and Cardiology while auditing unpaid insurance claim write-offs to raise realization rates above 80%.

10. Conclusions :

The integration of Excel for structured data pre-processing and Power BI for dynamic visualization established an end-to-end analytics pipeline that converted complex healthcare operational logs into strategic business intelligence. By identifying and uncovering an overall payment realization rate . This project equips healthcare administrators with the insights needed to optimize provider allocation, streamline billing workflows, and target claim realization recovery.